DIGITAL PUBLIC SAFETY PLATFORM

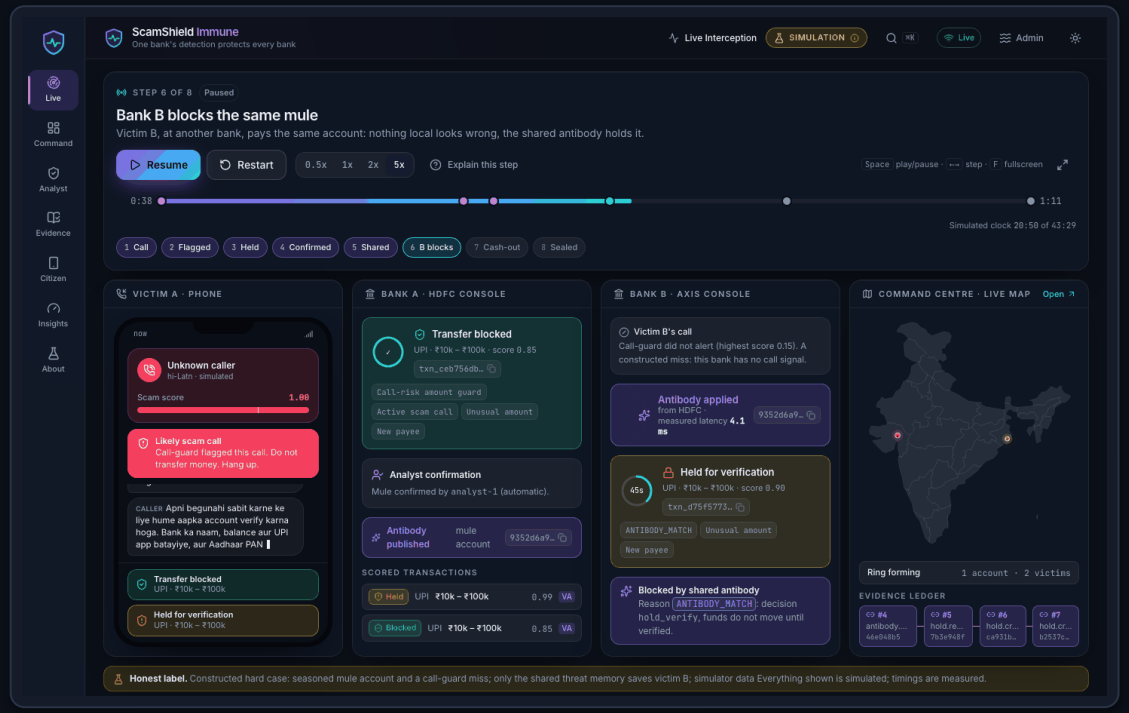
ScamShield Immune

One bank's detection protects every bank

Reflex, shared memory and a fraud-network brain, with evidence a third party can verify.

SIMULATED DATA

Benchmark seed 42, hero run seed 1



THE PROBLEM

Reactive investigation loses to scams that move in minutes

1.14 million

complaints, India, 2023

Rs 1,776 crore

digital-arrest scams, first nine months of 2024

Why it fails

Money leaves before any report.

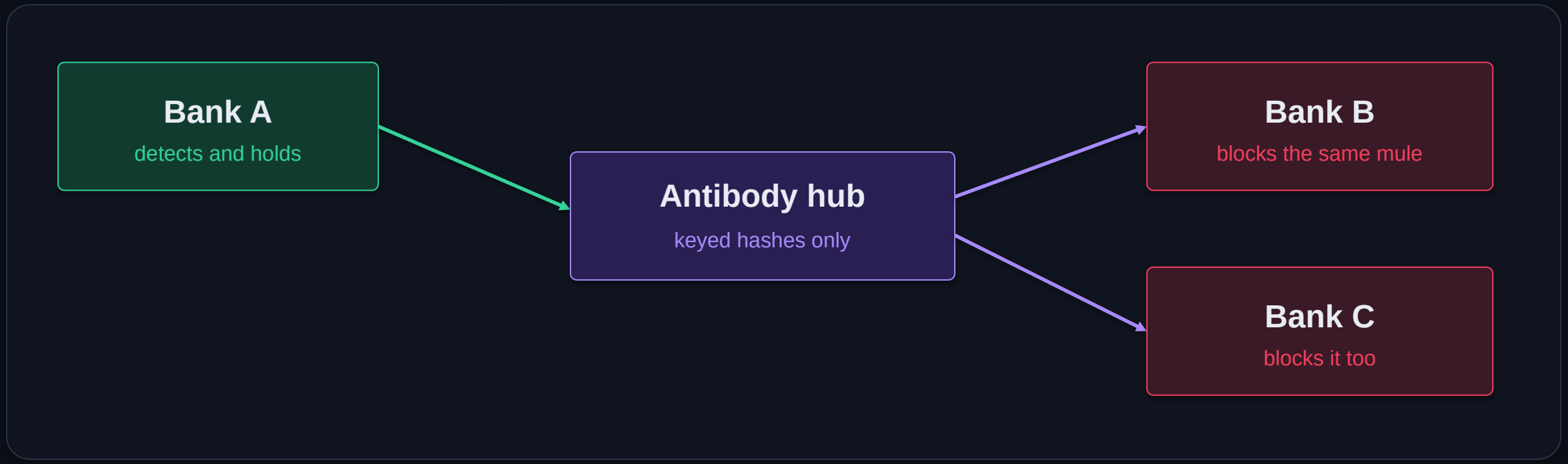
Banks see only their own customers.

Mules are reused on the next victim.

Source: challenge statement

THE INSIGHT

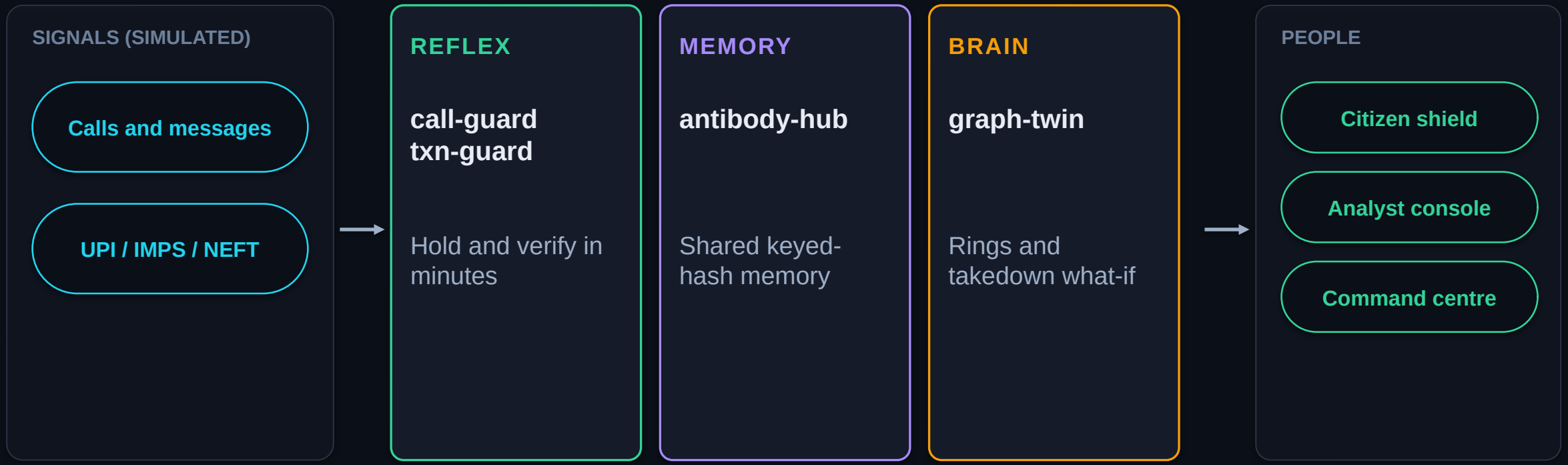
An immune system for payments



Detect once, protect everywhere. Hold and verify before the money moves.

WHAT WE BUILT

Three layers, a citizen shield and evidence you can verify

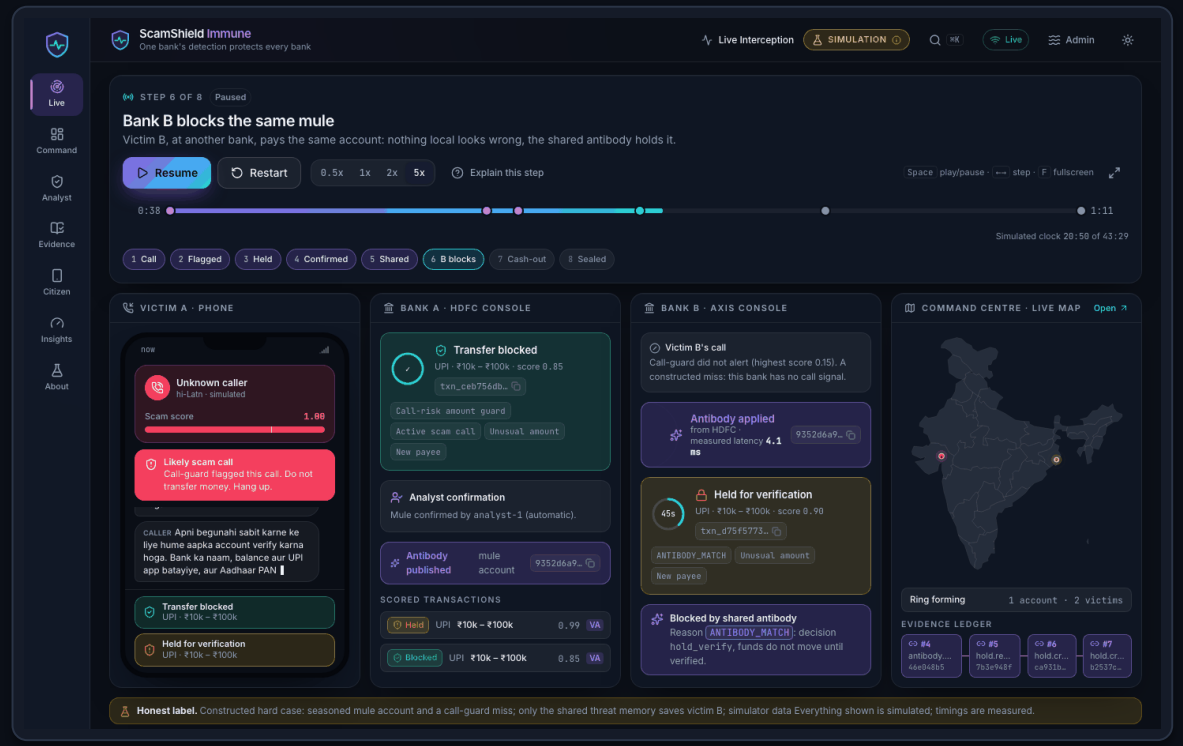


EVIDENCE LEDGER

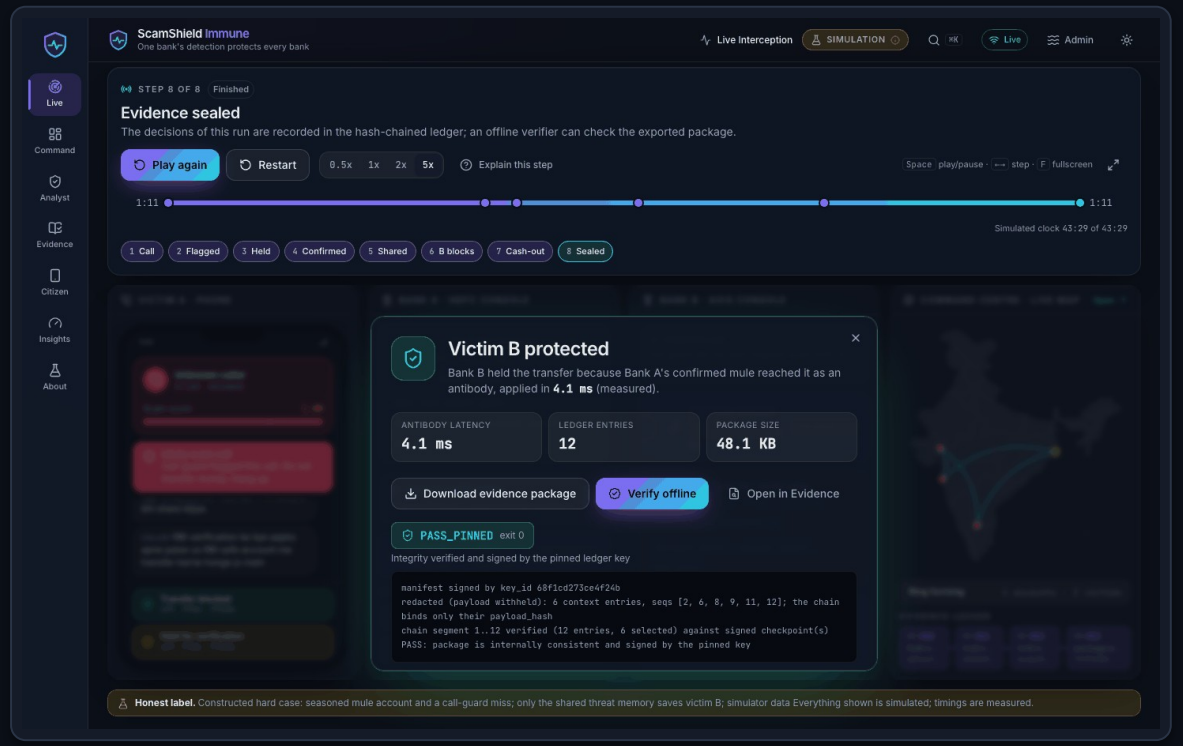
Hash chain, signed checkpoints, signed package, offline verify.py

THE HERO RUN

One constructed case, eight steps, 71 seconds at 1x



Bank B blocks the same mule



Victim B protected, evidence sealed

1 Call begins	2 Call flagged	3 Transfer held	4 Analyst confirms	5 Antibody shared	6 Bank B blocks	7 Cash-out	8 Evidence sealed

UNDER THE HOOD

Real services and models, a simulated world

Piece	Real	Simulated
Call scoring	clf-v1 classifier	scripts, speakers
Txn scoring	gbm-v1 model	accounts, transfers
Threat memory	hub store, outbox	two banks
Evidence	Ed25519 chain, verifier	simulated clock
Analyst	resolve API	60 s oracle

make demo runs the real services in one process.

RESULTS

Honest results: no false holds, with a weak NEFT rail

100.0%
Precision, hold
95% CI 99.3 to 100.0

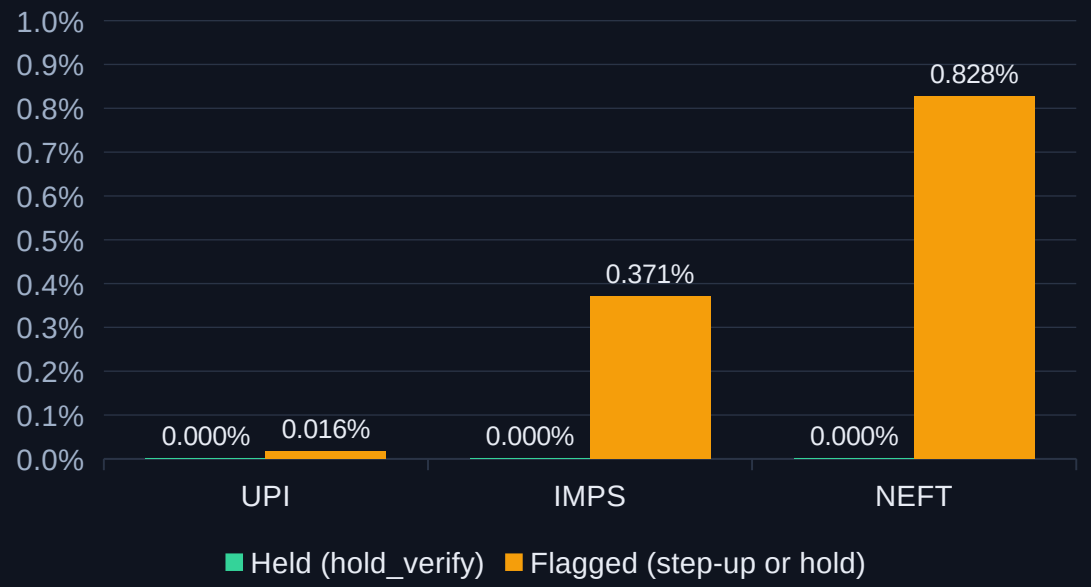
82.6%
Recall, scam txns
95% CI 79.6 to 85.3

0.000%
Benign held (FPR)
upper CI 0.018%

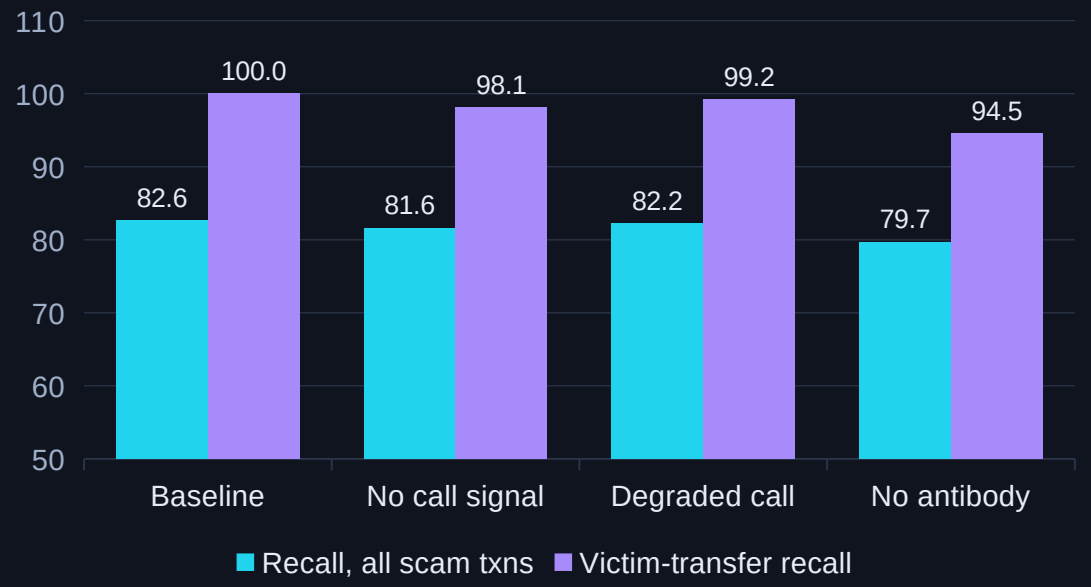
+6h 29m
Mean lead time
5 of 5 before 10th victim

Benign false-positive rate by rail (%)

NEFT: no scam data

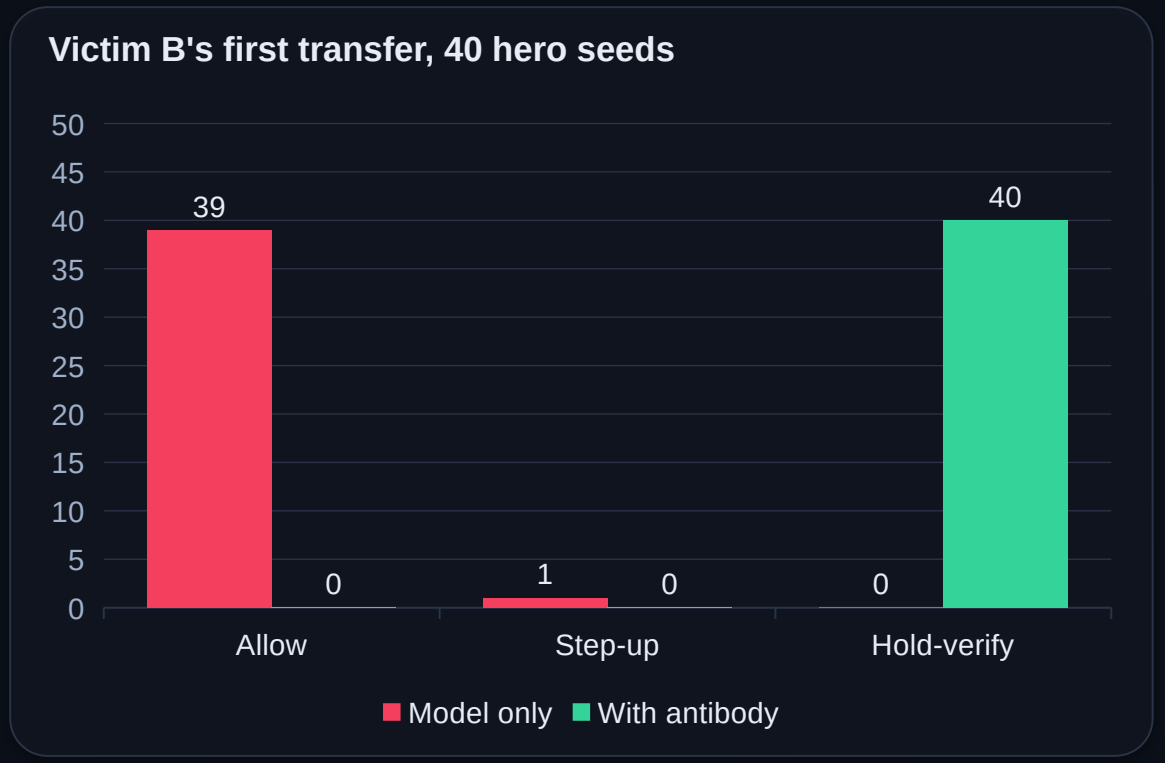


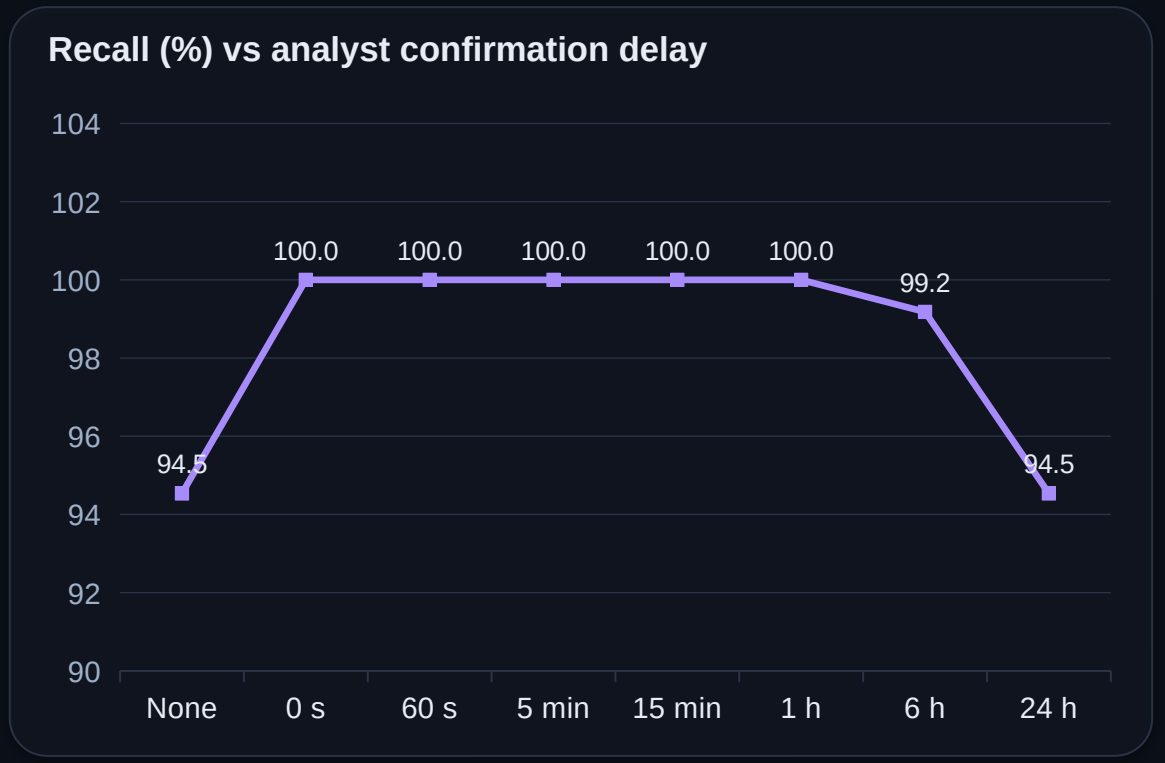
Ablations: recall (%), axis from 50



HARD-CASE HONESTY

Hard on purpose: antibodies must land early



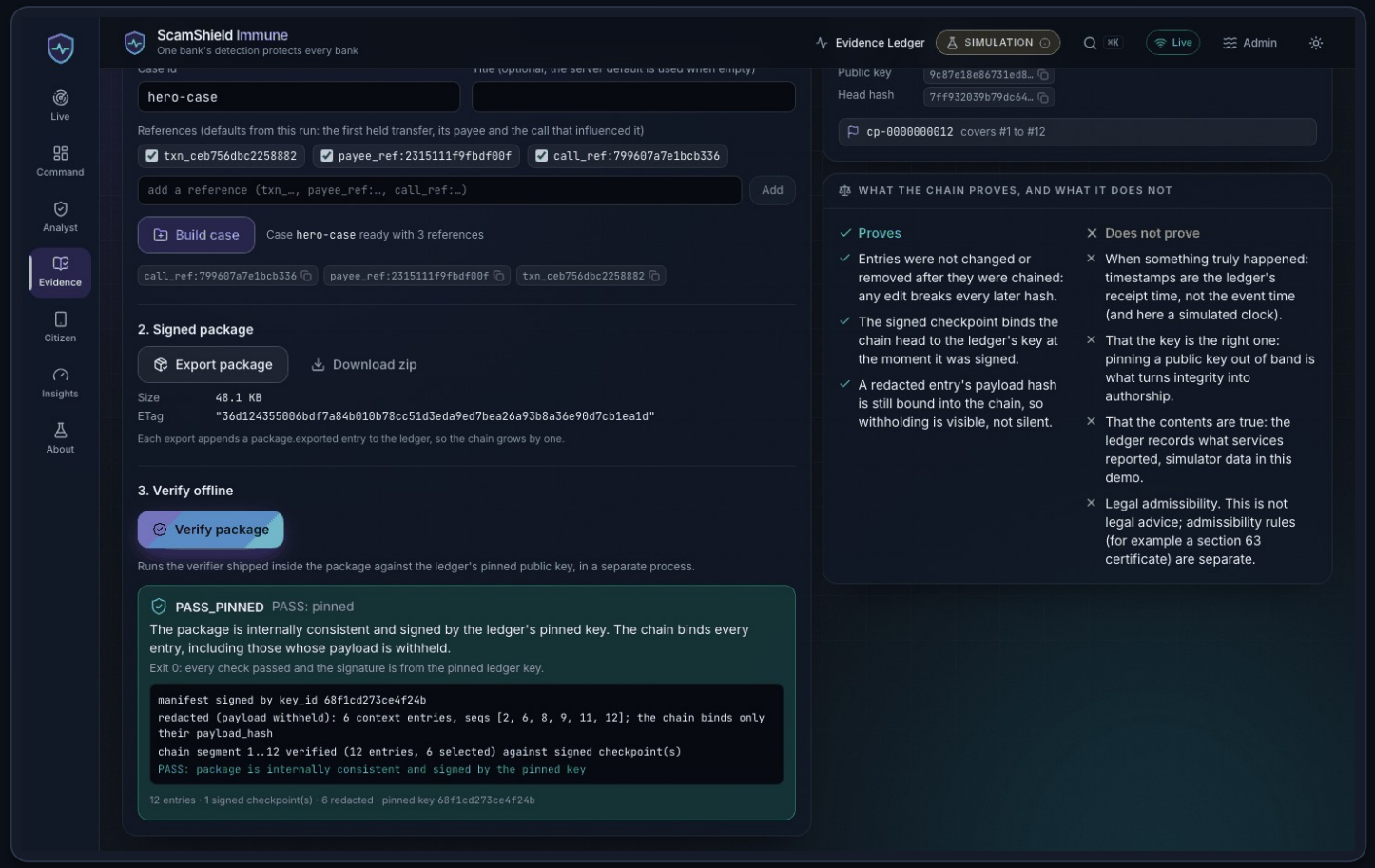


Seasoned mule, off-script call: the model alone allowed 39 of 40; with the antibody all 40 were held.

At a 24 h confirmation delay the benefit is gone (94.5%).

AUDITABILITY

Evidence a third party can check without trusting us



PROVES

Entries unaltered, in order, signed under a pinned key.

DOES NOT PROVE

That events were true, or legally admissible.

VERIFY OFFLINE

```
python -I -S verify.py --trusted-pubkey KEY
```

Payloads hold ids, reason codes and amount buckets only; PII-shaped entries are quarantined.

PRIVACY AND SAFETY ENGINEERING

Tokens and hashes only, and audit never blocks an alert

Tokens, not identities

Keyed hashes only.

PII guard

No amounts or free text.

Never on the hot path

Alerts publish before audit.

Fail safe

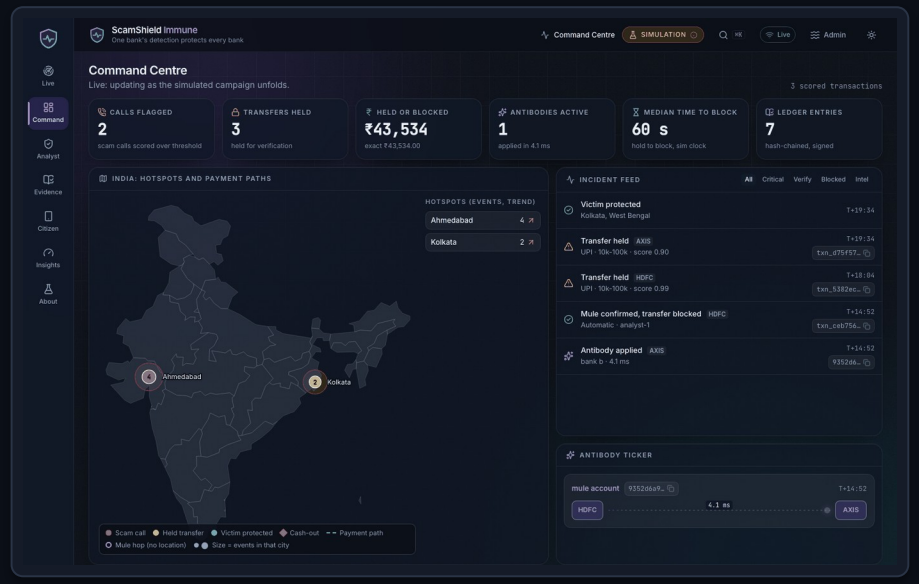
Rules fallback; no auto-release.

INDEPENDENT ADVERSARIAL REVIEW

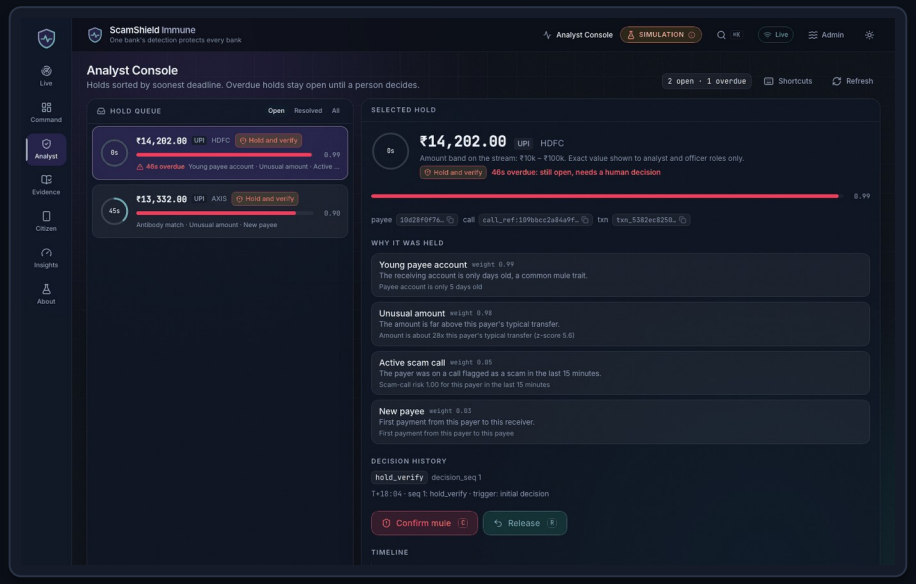
Independent adversarial review of each service; about 15,000 ledger mutations, none passed.

THREE AUDIENCES

Command centre, analyst console and citizen shield



Command centre: map, ring, takedown



Analyst: deadline-sorted hold queue



Citizen verdict, Tamil

English, Hindi, Tamil, Bengali

axe-core: 0 serious or critical

Keyboard shortcuts, palette

Risk is never shown by colour alone; one verdict per screen for citizens.

SCALABILITY AND OPERATIONS

Designed to scale by partition key; not load-tested here

Topic	Partition key
`txn.events`	payer_token
`call.events, call.risk`	victim_token
`txn.decisions`	txn_id
`antibody.published`	key_hash

7 ms

p50 per transaction

p99 about 14 ms

3

compose profiles

core, stack, observability

Audit outbox; stateless scorers.

Latency is in-process. The Docker stack was validated, not run.

What we do not know yet, and what comes next

LIMITS

- Simulator data only
- No NEFT scam data
- Modelled analyst
- Stack not run

NEXT STEPS

1. Real data integration
2. NEFT scam data
3. Graph-twin on live traffic
4. Legal review of evidence
5. One-bank pilot

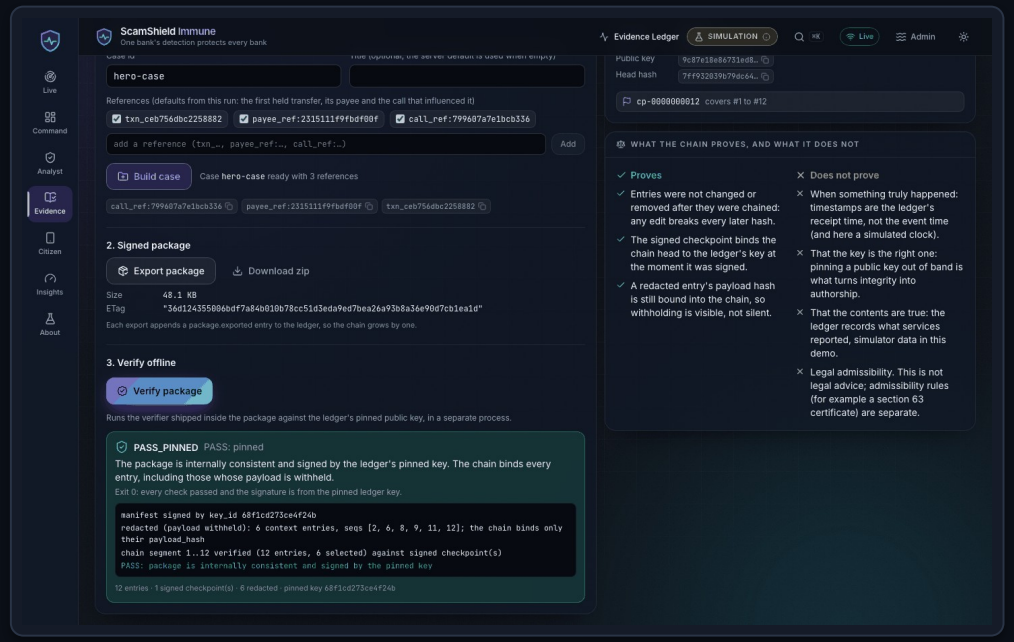
TRY IT

Run it in one command

```
make demo
```

Open http://localhost:8080, choose Officer, press Play.

github.com/aman-18-choudhary/ai_powered_immune (branch build/implementation)



Thank you

SIMULATED DATA